

		$g - (t_2^2 - t_1^2) \sin \theta$
4	A	Rate of change of momentum = $\frac{\Delta(mv)}{t}$ $= v_{rel} \frac{\Delta m}{t}$ $= v_{rel} \frac{(\rho \times \text{vol of water expelled})}{t}$ $= v_{rel} \frac{(\rho \times \pi r^2 h)}{t}$ $= v \rho \pi r^2 \frac{h}{t}$ $= v \rho \pi r^2 v$ $= \rho \pi r^2 v^2$ $= 1000 \times \pi \times 0.012^2 \times 10^2$ $= 45 \text{ N}$

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5	C	Assume Charlie throws the ball to the right and take rightwards as positive
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